

1. Business Task

Cyclistic is a bike-sharing company operating in Chicago.

Financial analysts have concluded that annual members are more profitable than casual riders. Therefore, the marketing department aims to increase the number of annual memberships.

The objective of this analysis is to identify behavioral differences between casual riders and annual members in order to support data-driven marketing strategies that encourage membership conversion.

Stakeholders:

- Lily Moreno – Director of Marketing
- Cyclistic Marketing Team
- Cyclistic Executive Team

2. Data Sources

The dataset used in this analysis comes from the publicly available historical trip data provided by Divvy/Cyclistic.

Twelve monthly CSV files were used, each representing one month of bike-sharing activity over the last year. Each row corresponds to a single bike trip.

The dataset includes variables such as:

- Ride ID
- Bike Type
- Ride start date and time
- Ride end date and time
- Start station
- End station
- Start coordinates
- End coordinates
- User type (casual and member)

The dataset does not contain any personally identifiable information (PII). Therefore, it is not possible to analyze users' age, gender, income, place of residence, individual identity, or whether a casual rider eventually became an annual member.

As a result, the analysis is limited to identifying observable riding patterns. Although these patterns may suggest certain user behaviors, they cannot confirm the actual purpose of each trip or the reasons why a casual rider would decide to purchase an annual membership.

3. Tools

The following tools were used throughout this project:

- **Google BigQuery:** Data import, integration, cleaning, transformation, and SQL analysis.
- **Power BI:** Data visualization, dashboard development, and presentation of insights.
- **Python (optional):** Used only when necessary to merge CSV files locally before uploading them to BigQuery due to file size limitations.

4. Data Cleaning and Preparation

The twelve monthly CSV files were imported into Google BigQuery and combined into a single table named `all_trips`.

A cleaned version of the dataset, named `clean_trips`, was then created. The cleaning process included:

- Removing rides with missing timestamps.
- Removing rides with negative or zero duration.
- Removing rides longer than 24 hours to eliminate obvious outliers.
- Creating additional variables to facilitate temporal analysis, including:
 - Ride duration
 - Day of the week
 - Hour of the day
 - Day type (Weekday/Weekend)
 - Season of the year

After the cleaning process, the dataset was reduced from 5,697,455 records to 5,535,677 valid rides. A total of 161,778 records (2.84%) were removed because they contained invalid values or unrealistic ride durations.

Data Cleaning Summary	Value
Original dataset	5697455 rides
Records removed	161778 rides
Percentage removed	2.84%
Final cleaned dataset	5535677 rides

Table 1. Data Cleaning Summary

```
1 CREATE OR REPLACE TABLE
2 'cyclistic-capstone-499723.Cyclistic.clean_trips'
3
4 AS
5
6 SELECT DISTINCT
7   ride_id,
8   rideable_type,
9   started_at,
10  ended_at,
11
12  TIMESTAMP_DIFF(
13    ended_at,
14    started_at,
15    MINUTE
16  ) AS ride_length_min,
17
18  DATE(started_at) AS ride_date,
19
20  EXTRACT(YEAR FROM started_at) AS year,
21
22  EXTRACT(MONTH FROM started_at) AS month,
23
24  EXTRACT(HOUR FROM started_at) AS hour,
25
26  EXTRACT(DAYOFWEEK FROM started_at)
27  AS day_number,
```

Figure 1. Fragment of SQL data cleaning process performed in Google BigQuery.

5. Analysis

The analysis compares casual riders and annual members using several behavioral metrics, including:

- Ride volume
- Average and median ride duration
- Weekday versus weekend usage
- Weekly riding patterns

- Hourly riding patterns
- Seasonal trends
- Bike type distribution
- Most frequently used stations

The objective is to identify significant behavioral differences between both user groups that could support future marketing strategies aimed at increasing annual memberships.

6. Visualizations

The findings of this analysis were presented through an interactive dashboard developed in Power BI. The dashboard was designed to provide a clear comparison between casual riders and annual members while highlighting the most relevant behavioral differences identified during the analysis.

The dashboard consists of four main sections:

Executive Summary

Provides an overview of the most important business metrics, including total ride volume, user distribution, average ride duration, and high-level usage trends.

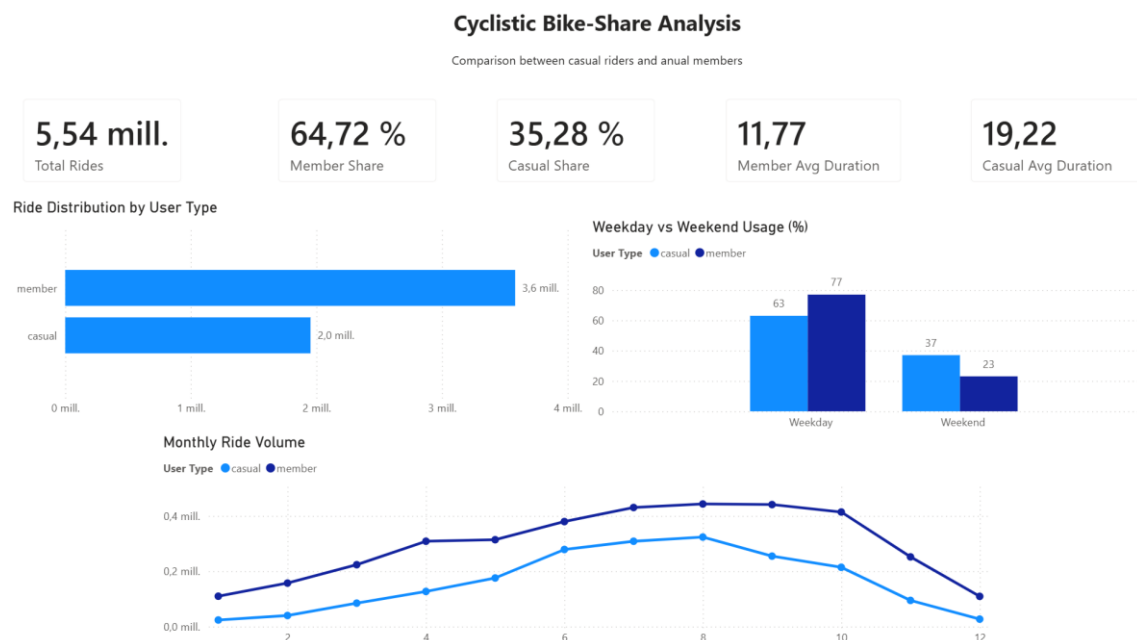


Figure 2. Executive Summary dashboard showing key business metrics, user distribution and monthly ride volume.

Temporal Behavior

Examines how riding patterns change over time by comparing hourly demand, weekday usage, and monthly ride duration between user groups.

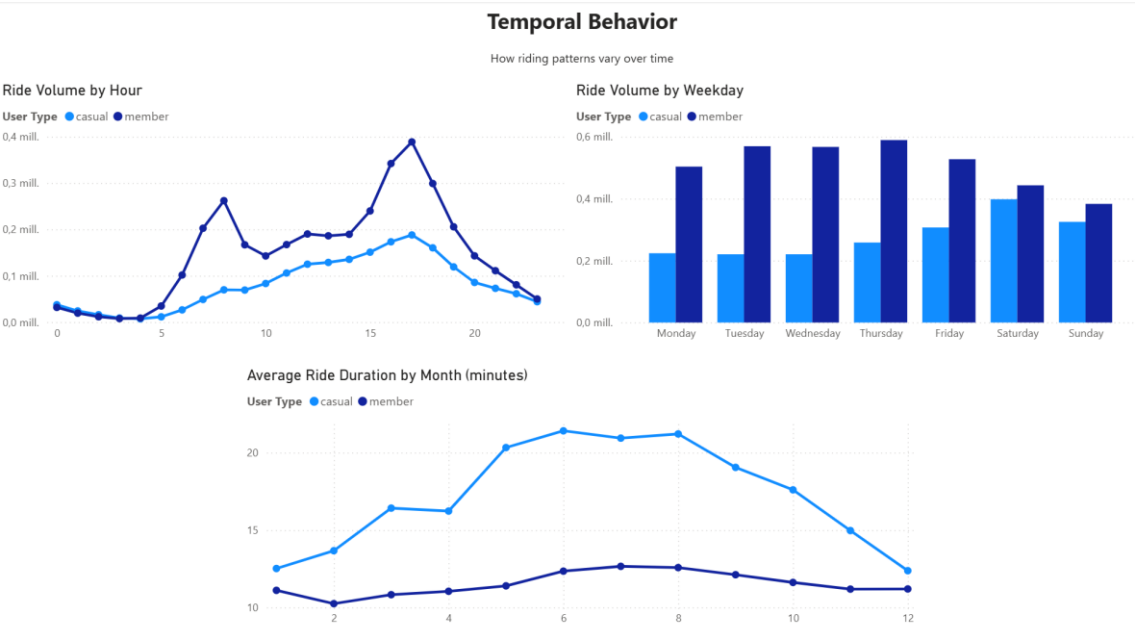


Figure 3. Temporal Behavior dashboard comparing hourly demand, weekday usage and average ride duration.

Spatial Analysis

Identifies the stations most frequently used by casual riders and compares bicycle type preferences between casual riders and annual members.

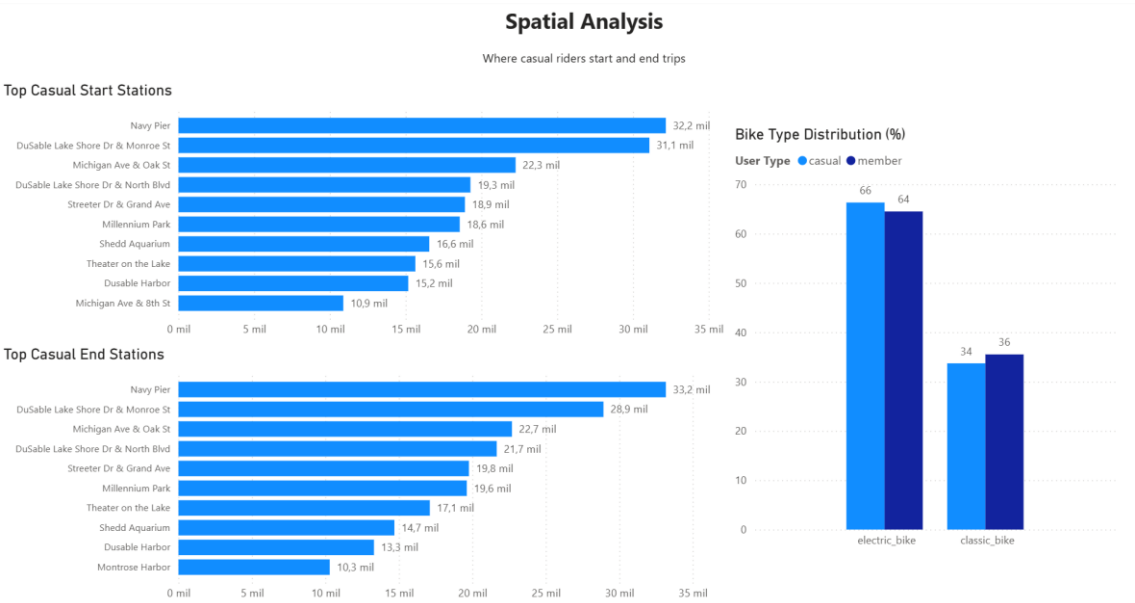


Figure 4. Spatial Analysis dashboard showing the most frequently used stations and bike type distribution.

Recommendations

Summarizes the main business insights and presents data-driven recommendations aimed at increasing the conversion of casual riders into annual members.

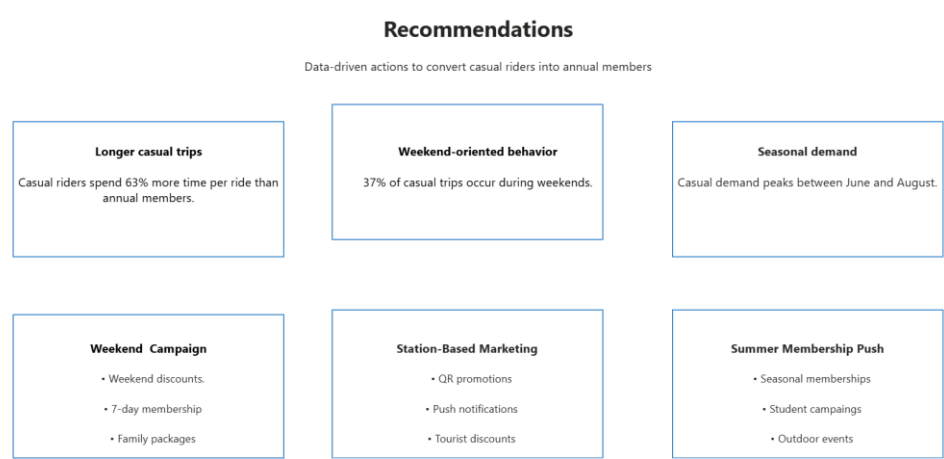


Figure 5. Business recommendations derived from the analytical findings.

7. Key Findings

User Distribution

Annual members generated 64.72% of all rides, while casual riders accounted for 35.28%.

Overall, annual members completed approximately 1.8 times more rides than casual riders, suggesting a much higher level of engagement with the bike-sharing service.

This finding reinforces the strategic importance of increasing the number of annual memberships, as members represent the customer segment with the highest riding frequency and the greatest long-term value for the company.

Ride Duration

The average trip duration for casual riders was 19.22 minutes, compared with 11.77 minutes for annual members.

Similarly, the median trip duration was 11 minutes for casual riders and 8 minutes for annual members.

Overall, casual riders spent approximately 63% more time per trip than annual members.

These results suggest that casual riders may use the service for leisure activities, tourism, or less routine journeys. However, because the dataset does not contain information about trip purpose, this interpretation should be considered a hypothesis based solely on observable behavioral patterns rather than a confirmed conclusion.

Weekday vs Weekend Usage

Among casual riders, 37.03% of all rides took place during weekends, whereas only 23.06% of annual member rides occurred on weekends.

Although annual members still generated a higher number of rides overall, casual riders showed a considerably greater proportional presence during weekends.

This pattern indicates that weekends represent a particularly valuable opportunity for marketing campaigns aimed at encouraging casual riders to purchase annual memberships.

Weekly Riding Patterns

A total of 76.94% of annual member rides occurred during weekdays, compared with 62.97% for casual riders.

Furthermore, annual members maintained relatively stable riding activity from Monday through Friday.

In contrast, casual riders showed a more concentrated usage pattern, with significantly higher activity during weekends.

These differences suggest that annual members rely on the bike-sharing service more consistently throughout the week, whereas casual riders appear to use the system more selectively.

Ride Duration by Day Type

Casual riders not only used the service more frequently during weekends but also completed considerably longer trips.

Their average ride duration increased from 17.56 minutes on weekdays to 22.05 minutes during weekends.

Annual members exhibited much smaller differences, averaging 11.41 minutes on weekdays and 12.93 minutes on weekends.

This finding further supports the hypothesis that casual riders are more likely to use the service for recreational activities, particularly during weekends.

Hourly Riding Patterns

Casual rider activity gradually increased from late morning and reached its highest levels between 12:00 PM and 5:00 PM.

Annual members, on the other hand, displayed a more consistent riding pattern throughout the day, with two distinct demand peaks between 7:00–9:00 AM and 4:00–6:00 PM.

These peaks are consistent with commuting-related travel, although the available data does not allow this explanation to be confirmed.

Overall, annual members demonstrated a more regular and evenly distributed riding pattern, while casual riders concentrated a larger proportion of their trips during the middle of the day.

Seasonal Trends

Casual rider activity reached its highest levels during the summer months and declined substantially during winter.

Annual members, by contrast, maintained a relatively stable level of usage throughout the year.

This stronger seasonality among casual riders suggests that the warmer months provide the best opportunity to launch targeted marketing campaigns focused on membership acquisition.

Bicycle Type Distribution

Electric bicycles were the most frequently used bike type for both user groups.

Among casual riders, 66.30% of all trips were completed using electric bikes, compared with 64.49% for annual members.

The popularity of electric bicycles across both customer segments suggests that they could be used as a central element in future promotional campaigns designed to encourage membership conversion.

Most Popular Stations

The start and end stations most frequently used by casual riders showed a highly consistent pattern.

The most popular locations included:

- Navy Pier
- DuSable Lake Shore Dr & Monroe St
- Michigan Ave & Oak St
- Millennium Park
- Shedd Aquarium
- Theater on the Lake

These stations are primarily located in recreational, tourist, and high-traffic urban areas.

The concentration of casual rider activity around these locations makes them ideal candidates for localized marketing initiatives, such as QR-code promotions, digital advertising, or limited-time membership offers.

8. Recommendations

Based on the results obtained throughout the analysis, several data-driven recommendations can be proposed to encourage casual riders to become annual members.

1. Launch Weekend-Focused Membership Campaigns

The analysis showed that casual riders have a significantly higher proportion of weekend trips than annual members. This indicates that weekends represent an ideal opportunity to promote annual memberships.

Cyclistic could implement targeted weekend campaigns, such as discounted memberships, limited-time offers, or promotional events, aimed specifically at casual riders during periods of peak recreational demand.

2. Promote Memberships During the Summer Season

Casual rider activity reaches its highest levels during the summer months, while annual members maintain relatively stable usage throughout the year.

Marketing campaigns launched between June and August are therefore likely to reach the largest number of potential customers.

Seasonal promotions, outdoor events, and partnerships with local attractions could increase membership visibility during this period of high demand.

3. Implement Station-Based Marketing Strategies

The analysis identified several stations with consistently high casual rider activity, including Navy Pier, Millennium Park, and Shedd Aquarium.

These locations provide excellent opportunities for localized marketing initiatives, such as:

- QR codes linking directly to membership registration.
- Digital advertising displayed near docking stations.
- Tourist-oriented promotional campaigns.
- First-month membership discounts for new subscribers.

Concentrating marketing resources at these high-demand locations could improve campaign efficiency while maximizing exposure among potential customers.

4. Promote the Benefits of Annual Memberships for Recreational Riders

Casual riders spend approximately 63% more time per trip than annual members.

Future marketing campaigns should emphasize the benefits that annual memberships provide for longer recreational rides, including unlimited ride access, lower long-term costs, and greater flexibility for frequent users.

Communicating these benefits clearly may help casual riders recognize the financial advantages of becoming annual members.

9. Limitations

Although this analysis provides valuable insights into customer behavior, several limitations should be considered when interpreting the results.

First, the dataset contains only historical trip records and does not include any demographic information such as age, gender, occupation, income, or place of residence. Consequently, it is not possible to identify which population groups are most likely to purchase an annual membership.

Second, the dataset does not provide information about the purpose of each ride. Although commuting and leisure patterns can be inferred from temporal trends, these interpretations cannot be confirmed using the available data.

Another important limitation is that the analysis covers only one year of historical data. Longer time series could reveal additional long-term trends or changes in customer behavior.

Finally, the dataset does not track individual customer journeys over time. Therefore, it is impossible to determine whether a casual rider eventually became an annual member or to directly measure conversion rates.

These limitations should be taken into account when developing future marketing strategies based on the findings presented in this report.

10. Conclusion

This project analyzed more than 5.5 million bike trips to identify behavioral differences between casual riders and annual members of Cyclistic's bike-sharing service.

The results demonstrate that annual members use the service more frequently and exhibit consistent riding patterns throughout the week, suggesting that cycling is integrated into their daily routines.

In contrast, casual riders complete fewer rides but spend considerably more time on each trip. Their riding activity is concentrated during weekends, increases substantially during the summer months, and is primarily associated with recreational and tourist-oriented locations.

These behavioral differences provide valuable opportunities for targeted marketing initiatives. Weekend promotions, seasonal campaigns, and station-based

advertising at high-demand locations could increase the likelihood of converting casual riders into annual members.

Overall, this analysis demonstrates how data can support strategic business decisions by identifying customer behavior patterns and translating analytical findings into practical recommendations. By combining SQL, Google BigQuery, and Power BI, the project illustrates a complete end-to-end data analysis workflow, from data cleaning and transformation to visualization and business decision-making.